

# MB Note

## 1 HIV/SIR Model

$$\text{HIV: } \begin{cases} \dot{x} = \lambda - dx - \beta xv \\ \dot{y} = \beta xv - ay \\ \dot{v} = ky - uv \end{cases} \Rightarrow$$

$\lambda$ : formation rate of healthy cells.  
 $d$ : death rate of healthy cells.  
 $\beta$ : infection rate.  
 $a$ : death rate of infected cells.  
 $k$ : formation rate of free viruses.  
 $u$ : death rate of free viruses.

$\hat{v}$  中缺  $-\beta xv$  的原因:  
The model neglects the cost of virus when it infects healthy cells.

**virus-free steady-state:**  $y = 0$   
and  $v = 0$ , possible  $x = \frac{\lambda}{d}$ .

**Estimate virus increase: in each virus generation/over the characteristic lifetime:**

Have:  $(x, y, v)(0) = (\frac{\lambda}{d}, 0, v_0)$ . 1. Period of free virus lifetime:  $u^{-1}$ . Period of infected cell lifetime:  $a^{-1}$ .

$$2. \frac{1}{u} \ll \frac{1}{a} \quad 3. v(\frac{1}{u}) - v(0) = \int_0^{\frac{1}{u}} \dot{v} dt = k \int_0^{\frac{1}{u}} y dt - u \int_0^{\frac{1}{u}} v dt$$
$$a. \int_0^{\frac{1}{u}} y dt \approx \int_0^{\frac{1}{u}} y(\frac{1}{a}) dt \quad b. \int_0^{\frac{1}{u}} v dt \approx \int_0^{\frac{1}{u}} v(0) dt, \quad a. y(\frac{1}{a}) = \int_0^{\frac{1}{a}} \dot{y} dt \approx \int_0^{\frac{1}{a}} \dot{y}(0) dt$$

**Basic Reproductive ratio:**  $R_0 = \frac{k\beta\lambda}{a d u}$ .  $R_0 > 1$  virus invades;  $R_0 < 1$  virus dies out.

估计 (e.g.  $R_0$ ) 某个值范围 | 最后趋近于某个值 | 得到趋近速率: 1. 计算 equilibrium. 2. 计算 Jacobian 算 eigenvalues. 3. 取最大 eigenvalue  $\lambda \rightarrow$  得 sol  $\propto e^{\lambda t}$ . 4. 根据题意估计求值范围 | 趋近值 | 速率 ( $\lambda$ ).

(e.g.) 证明  $u = \text{cont. when } t = \text{cont. } (u^{-1})$ : 1. Linearise 估算趋近值 2. 假设原式  $\approx$  估计值 3. 解出  $t$  估计

$$\text{SIR: } \begin{cases} S' = -\beta SI \\ I' = \beta SI - \alpha I \\ R' = \alpha I \end{cases} \Rightarrow \begin{cases} S: \text{susceptible individuals (易感人群).} \\ I: \text{infected individuals (感染者).} \\ R: \text{recovered individuals (康复者).} \end{cases}$$

$\beta$ : infection rate.  
 $\alpha$ : recovery rate. (or death, if no R)  
ps:  $\lim_{t \rightarrow \infty} I(t) = 0$   $\lim_{t \rightarrow \infty} S(t) > 0$  and  $R(0) = 0$

**Final Size Relation:**  $\ln(S_0/S_\infty) = \frac{\beta}{\alpha} [N - S_\infty] = R_0(1 - \frac{S_\infty}{S_0})$   
**proof:** 1.  $(S + I)' = -\alpha I$  2.  $\int_0^\infty (S + I)' dt = -\alpha \int_0^\infty I dt \Rightarrow (S + I)_\infty - (S + I)_0 = -\alpha \int_0^\infty I dt$   
3. By  $N = S_0 + I_0$ , can get  $S_\infty - N = \ln(S_\infty/S_0) \frac{\beta}{\alpha}$ .  $\square$

## 2 Relaxation/Excitation/Switch - FitzHugh Nagumo Model

**Explain why oscillations/relaxation occur:**

1. (e.g.)  $\dot{x} = c[f(x) - y]$  and  $\dot{y} = a$  line, where  $f(x)$  is a cubic-like curve.  $c \gg 1$   
2. show  $|\dot{x}| \gg |\dot{y}|$  (or vice versa), and no stable equilibrium exists. (can unstable)  
3. 图上标一起始点  $a$ , 画类似右侧图。 4. (e.g.) a. Start at  $a$ ,  $|\dot{x}| \gg |\dot{y}|$ , so it quickly moves horizontally to the right branch of cubic-like curve (A). b. When it closed to the right branch, since  $\dot{y} < 0$ ,  $\dot{x} \approx 0$ , so it moves downwards along the right branch. Then, it closes to the point B. It has option to leave the path  $f(x)$  since  $\dot{y} < 0$ , so it cannot follow  $f(x)$  and again quickly jumps from B to the left branch of the cubic-like curve (C) since  $|\dot{x}| \gg |\dot{y}|$ . When it closed to the left branch, since  $\dot{y} > 0$ ,  $\dot{x} \approx 0$ , so it moves upwards along the left branch. Finally, it reaches the point D, and again quickly jumps to the right branch of the cubic-like curve (A). This process repeats, resulting in relaxation oscillations.

**Estimate Period:** (neglect fast jumps) 1. Period  $T \approx \int_t^t \frac{dy}{dx} dt + \int_C^t \frac{dx}{dy} dt = 2 \int_t^t \frac{dx}{dy} dt$  (by symmetry)

$$2. (e.g.) \dot{x} = c[f(x) - y], \dot{y} = g(x, y) \text{ then } y \approx f(x), \frac{dy}{dx} \frac{dx}{dt} = f'(x) \frac{dx}{dt} \Rightarrow dt = \frac{f'(x)}{g(x, y)} dx$$

3. Find coordinates of A, B, C, D (turning points)  $\Rightarrow T \approx 2 \int_A^x B - dx$

**Three kinds of Excitable System:**

**small perturbation:** returns to equilibrium. (fail excitation)

**large perturbation:** large excursion in phase space before returning to equilibrium. (action potential/successful excitation)

**Refractory period**(不应期): system insensitive to further

**constant stimulus:** periodic excitation. (repetitive firing/limit cycle)

**Explain why excitable occur:** 结合前面 relaxation/oscillation 的分析, stable equilibrium (图为原点), 分析 P/A 开始 (small|large perturbations).

**FitzHugh-Nagumo Model:**

$$\text{Have: } \begin{cases} \dot{v} = f(v) - w + I(t) \\ \dot{w} = bv - \gamma w \end{cases} \Rightarrow \begin{cases} v: \text{membrane (excitable) potential.} \\ w: \text{ionic current. (recovery var)} \end{cases}$$

$a, b, c$ : positive constants. **I(t)**: external stimulus.

**Switch Behaviour:** 类似 relaxation oscillation, 有两个 stable equilibria, 分析一个过去后不会回来。

**Remark:** 对于这类模型, 计算平衡点用  $f(v)$  的图像斜率和另一条直线斜率来判断。

## 3 Stability/Linearise Stability

**Continuous**  $\frac{dN}{dt} = f(N)$ :

- **Stable:**  $f'(N^*) < 0$

- **Unstable:**  $f'(N^*) > 0$

**Discrete**  $N_{t+1} = f(N_t)$ :  $\lambda = f'(N^*)$

- **Stable:**  $|\lambda| < 1$  - **periodic:**  $|\lambda| = 1$

- **Unstable:**  $|\lambda| > 1$  (chaotic)

- **tangent bifurcation:**  $\lambda = 1$

- **period-doubling bifurcation:**  $\lambda = -1$

**One var.  $\uparrow$  | More var.  $\downarrow$ :**

**Jacobian:**  $J = [F_x \ F_y \ \dots]$

**Eigenvalue Method:**

If  $\text{Re}(\lambda_i) < 0$  for all  $i$ , then asy stable.

If  $\lambda_1 < 0 < \lambda_2$ , then saddle (unstable).

If  $\text{Im}(\lambda_i) \neq 0$ , then focus (spiral).

If  $\text{Re}(\lambda_i) = 0$ , then center (linear!).

or (nonlinear): **Hopf bifurcation.**

If  $\lambda = 0$ : If  $\lambda_1 = 0, \lambda_2 < 0$ : It's a "saddle-node" with one stable eigendirection (corresponding to  $\lambda_2 < 0$ ) and one "center" eigendirection (corresponding to  $\lambda_1 = 0$ ). ps: 如果  $\lambda > 0$  或都 = 0, 类似分析。

**Linearise Stability:** 1. Let  $N(t) = N^* + n(t), |n(t)| \ll 1$ . or  $N_t = N^* + n_t, |n_t| \ll 1$ .

$$2. \frac{dn}{dt} = \frac{dN}{dt} = a. f(N^*) + f'(N^*)n(t) \text{ or } f(N^*) + f'(N^*)n_t \quad b. (\text{delay}) \text{ 直接代入后 linearise.}$$

**Justify bifurcation:** 1. Assume  $n(t) = e^{\lambda t}$ . 2. 代入  $\frac{dn}{dt}$ :  $\lambda e^{\lambda t} = f'(N^*)e^{\lambda t}$  3. (Re), (Im) 得两方程。

4a. 反证  $\Re(\lambda) > 0, < 0$  or **4b.find**  $T_c$ . Check  $T = 0$  stable/unstable  $\Rightarrow$  first time  $\Re(\lambda) = 0$

**Properties for continuous: Time-Independent:** At  $N^*, N(t - T) = N(t) = N^*$ .

**Tangency-Condition:** At bifurcation point  $(x^*, r_c), f_x(x^*, r_c) = f(x^*, r_c) = 0$ .

**Properties for discrete: eigenvalue:**  $\lambda = f'(N^*)$ .

(for perturbations)  $\lambda < -1$ :  $\uparrow$  oscillation;  $-1 < \lambda < 0$ :  $\downarrow$  oscillation;

$0 < \lambda < 1$ :  $\downarrow$  monotonic;  $\lambda > 1$ :  $\uparrow$  monotonic.

**p-periodic:**  $N^* = f^p(N^*)$ . Every  $p$ -periodic  $N^*$  with  $\lambda = -1$  branches into a  $2p$ -periodic  $N^*$ .

**Sarkovskii's Theorem:** If a 3-periodic  $N^*$  exists, then  $p$ -periodic  $N^*$  exists for all  $p \in \mathbb{N}^+$ .

**Max/Min:**  $f'(N_m) = 0, N_{\max} = f(N_m), N_{\min} = f(N_{\max})$ . assume *unimodal*, 2-periodic exists.

**Types of Bifurcations: Saddle-Node:**  $\dot{x} = r \pm x^2$  (Normal Form)  $0 \rightarrow S \rightarrow U-S$

**Transcritical:**  $\dot{x} = rx - x^2$  (Normal Form)  $S-U \rightarrow U-S$

**super-critical Pitchfork:**  $\dot{x} = rx - x^3$  (Normal Form)  $S \rightarrow S-U-S$

**sub-critical Pitchfork:**  $\dot{x} = rx + x^3$  (Normal Form)  $U \rightarrow U-S-U$

**Hopf:** (平衡点从稳定变为不稳定, 产生 limit cycle) **Condition:**  $\det(J) > 0, \frac{d\text{tr}(J)}{d\mu} \neq 0, \mu$ : parameter, not  $t$

**or:** 找到  $\mu$  s.t.  $\text{tr}(J) > 0, \mu > \mu_c$  and  $\text{tr}(J) < 0, \mu < \mu_c$ . **Period:**  $T = \frac{2\pi}{\omega}$ , 特征向量:  $\lambda = \pm i\omega$

**Bifurcation Diagram:** x-axis: equilibrium value; y-axis: parameter value. 虚线: unstable; 实线: stable

**# ODEs**

$$y' + py = q: \text{Integrating Factor: } I = e^{\int p dt} \text{ sol: } I(x)y = \int I(x)q dt$$

$$y'' - \lambda y = 0: 1. \lambda > 0: y = c_1 \sinh(\sqrt{\lambda}t) + c_2 \cosh(\sqrt{\lambda}t) = c_1' e^{\sqrt{\lambda}t} + c_2' e^{-\sqrt{\lambda}t}$$

$$2. \lambda = 0: y = c_1 t + c_2 \quad 3. \lambda < 0: y = c_1 \sin(\sqrt{-\lambda}t) + c_2 \cos(\sqrt{-\lambda}t)$$

$$ay'' + by' + cy = 0: 1. \Delta > 0: y = c_1 e^{r_1 t} + c_2 e^{r_2 t} \quad r_{1,2} = \mu + vi$$

$$2. \Delta = 0: y = e^{\mu t}(c_1 + c_2 t) \quad 3. \Delta < 0: y = e^{\mu t}(c_1 \cos vt + c_2 \sin vt)$$

## 4 Nullclines/Limit Cycle/Trapping Region

**Nullclines:** For:  $u' = f(u, v), v' = g(u, v)$ .  $u$ -nullcline:  $f(u, v) = 0$ .  $v$ -nullcline:  $g(u, v) = 0$ .

**Slope of Nullclines:** Since  $df = f_u du + f_v dv = 0$  on  $u$ -nullcline,

Slope of  $u$ -nullcline:  $\frac{dv}{du} = -\frac{f_u}{f_v}$ . Similarly,  $\frac{dv}{du} = -\frac{g_u}{g_v}$  for  $v$ -nullcline.

**Trapping Region:** 1. 直线/nullcline 围闭合区域。 2. 边界上, 外法向量  $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

**Poincaré-Bendixon:** One unstable equilibria (no saddle) + trapping region = stable limit cycle exists.

**Useful Tools Polar Coord.:**  $rr' = xx' + yy'$  and  $r^2\theta' = xy' - yx'$ .  $\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$

## 5 Diffusion/Wave Model

**Diffusion Eq.:**  $u(x, t)$  s.t.  $u_t = D\Delta u$  with  $u(x, 0) = f(x)$   $D$ : diffusion coeff.

**Fundamental sol:**  $\phi(x, t) = \frac{Q}{(\sqrt{4\pi Dt})^{n/2}} e^{-\frac{|x|^2}{4Dt}}$ . If  $f(x) = Q\delta(x)$ ,  $u(x, t) = \phi(x, t)$

**General sol:**  $u(x, t) = \int_{\mathbb{R}^n} \phi(x - y, t) f(y) dy$

**Non-Homo:** If  $u_t \rightarrow u_g = D\Delta u + a(t)u$ , consider  $u(x, t) = v(x, t)e^{\int_0^t a(s)ds}$  with  $v_t = D\Delta v$ .

**Flux (J):** 1. 考虑  $u, J$  for continuous 函数. (尤其是分段函数) 2.  $J$  考虑物理意义, 边界上为 0.

e.g.  $J = -\mu b_x + \delta b(\ln s)_x$   $\mu$ : diffusion coeff.  $\frac{\delta}{\mu}$ : chemotactic sensitivity

**Interpret Initial val.  $u_0\delta$ :** Assumption that the {species} were released at a single point and nowhere else.

**Give initial condition:** Usually using  $u(x, 0) = u_0$  (uniform distribution)

**Give boundary condition:** Using  $u, J$  is continuous at boundary.  $\Rightarrow$  考虑边缘要求连续得到条件

**Wave Model:** Consider  $u(x, t) = u(x - ct) = U(z)$  with  $z = x - ct$   $c$ : wave speed

**Combaine diffusion:**  $u_t = Du_{xx} + f(u) \Rightarrow$  ODE: (换元后得到)  $-cU' = DU'' + f(U)$

**ODEs:** (Let  $V = U'$ )  $\begin{cases} U' = V \\ V' = -\frac{c}{D}V - \frac{f(U)}{D} \end{cases} \Rightarrow$  If heteroclinic: If wavefront: (or)  $\begin{cases} U(\pm\infty) = u^\pm \\ (u^\pm, 0) \text{ are equilibria of ODEs} \end{cases}$

**Homoclinic:**  $U(\pm\infty) = u^*$  (i.e.  $u^+ = u^-$ ) **ahead of wave:**  $u^-$  **behind wave:**  $u^+$

**Condition if wavefront:**  $\lim_{z \rightarrow \pm\infty} U(z) = u^\pm$  and  $\lim_{z \rightarrow \pm\infty} U'(z) = 0$

**Show the existence of wavefront|connection|heteroclinic between  $u^-$  and  $u^+$ :**

1. 确定  $u^\pm$ , 题目会给, 或由  $f(u)$  的零点确定。 2. 计算 Jacobian, 确定  $u^\pm$  的类型且计算 eigenvalues.

3. (ps: for  $J = [0, 1; a, b]$  have eigenvectors:  $[1; \lambda]^\top$ ) The positive|negative eigenvalues at  $u^-|u^+$  corresponding to an eigenvector  $[1; \lambda]^\top$ , which means that it has an unstable|stable eigendirection with positive|negative slope. 4. Therefore, it's plausible

**Wave Speed:** 1. For  $-cU' = DU'' + f(U)$  两边乘  $U'$  对  $z$  在  $(-\infty, \infty)$  积分  $= 0$

$$2. -c \int_{-\infty}^{\infty} (U')^2 dz = D \int_{-\infty}^{\infty} U'' U' dz + \int_{-\infty}^{\infty} f(U) U' dz \text{ by } U' dz = U' dz, \int_{-\infty}^{\infty} U'' U' dz = \frac{1}{2} [(U')^2]_{-\infty}^{\infty}$$

$$3. c = \left[ \int_{u^-}^{u^+} f(U) dU \right] / \left[ \int_{-\infty}^{\infty} (U')^2 dz \right] \text{ Remark: Sign of } c \text{ depends on area between } f(U) \text{ and } U\text{-axis}$$

**Meaning: critical front speed:**  $c = 0$  (对于灾害蔓延);  $c < 0 < c > 0$ : no spread(dies out)|spread(occurs)

**Possible  $c = 0$ :** Assume continuous dependence of  $c$  on {parameter  $p$ }. 证明有  $c > 0 < c < 0 \Rightarrow \exists p_c$  s.t.  $c = 0$

**Fisher-KPP:**  $u_t = Du_{xx} + ru(1 - \frac{u}{K}) \Leftrightarrow -cU' = DU'' + ru(1 - \frac{u}{K})$ , wavefront s.t.  $c \geq c_{\min} = 2\sqrt{rD}$ .

**Condition for Wavefront sol:** eigenvalue|separation by var. 算出来后 NO oscillation sol. (i.e. NO sin/cos)

**Remark:** 如遇计算 parameter 值 | 计算待定系数 | 算 wave speed sign: 直接积分 or 乘  $g, U$  分部积分!

## 6 Lotka-Volterra Model — Predator-Prey Model

**Model:**  $\begin{cases} \dot{x} = x(a - by) \\ \dot{y} = y(c - d) \end{cases} \Rightarrow \begin{cases} x: \text{prey pop.} & a: \text{prey growth rate.} & b: \text{predation rate coeff.} \\ y: \text{predator pop.} & c: \text{predator growth rate coeff.} & d: \text{predator death rate.} \end{cases}$

**Constant of Motion:** 1. 用  $\frac{dy}{dx} = \frac{y}{x}$  算出 trajectory 方程 e.g.  $H(u, v) = \alpha u + v - \ln(u^\alpha v) = H_0$

2. If  $\frac{dH}{dt} = 0$  or  $\frac{dH}{dt} = 0, H$  is constant of motion.

**Show trajectory closed:** a. Calculate  $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix}$  at equilibrium  $(u^*, v^*)$ .

b. If positive|negative definite  $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix} > 0|< 0$ , then  $H$  strictly convex|concave, has global min|max, closed.

c. Therefore, trajectory is closed. **Closed trajectory  $\Rightarrow$  limit cycle.** (periodic sol)

**Why periodic occurs:** As any trajectory must lies on one of the closed level curves of  $H$ , and we have shown that it is closed. Thus, the periodic oscillation occurs.

**Principal of Competitive Exclusion:**

If two species compete for the same limited resource, one will outcompete and exclude the other. ps: 图 b,c,d 都是。

**Competition Model:**  $\begin{cases} \dot{x} = x(r_1 - b_1 x - c_1 y) \\ \dot{y} = y(r_2 - b_2 y - c_2 x) \end{cases} \Rightarrow \begin{cases} x, y: \text{pop. of species 1, 2} \\ b_i: \text{intraspecific competition} \\ c_i: \text{interspecific competition} \end{cases}$

右图描述: (a): Stable coexistence. (Weak competition)

intraspecific competition > interspecific competition

(b): Bistability. (Strong competition, by initial condition 决定)

interspecific competition > intraspecific competition

(c), (d): One species outcompetes the other. interspecific competition > intraspecific competition

(e): Weak mutualism. Both species tend to a stable coexistence, benefiting from each other

(f): Strong mutualism. Due to absence of attractor, populations grow without bound, resulting population explosion.